$\vec{B} \perp \vec{k}$ $\quad \vec{B}_0 \neq 0$.

Case 1. $B_0 \parallel \vec{E}_1$

$$\vec{B}_0 = B_0 \hat{z} \quad \vec{k} = k\hat{x} \quad \vec{E}_1 = E_1 \hat{z}$$

Maxwel: $\quad \nabla \times \vec{B} = \frac{1}{c^2} \frac{\partial}{\partial t} \vec{E} + \mu_0 \vec{j}$

$$\nabla \times \vec{E} = -\frac{\partial}{\partial t} \vec{B}$$

$$\nabla \times (\nabla \times \vec{E}) = -\frac{\partial}{\partial t} (\nabla \times \vec{B}) = -\frac{1}{c^2} \frac{\partial^2}{\partial t^2} \vec{E} - \mu_0 \frac{\partial}{\partial t} \vec{j}$$

$$i\vec{k} \ (i\vec{k} \cdot \vec{E}) - (ik \ ik \vec{E}) = -(-i\omega) \left[ \frac{1}{c^2} (-i\omega \vec{E}) + \mu_0 \vec{j} \right].$$

$$\vec{j}_1 = -en_0 \vec{v}_{e1}$$

$$m \frac{\partial}{\partial t} \vec{v}_{e1} = -e (\vec{E}_1 + \vec{v}_1 \times \vec{B})$$

寻常波.

Case 2. $\vec{B_0} \perp \vec{E_1}$

$$\vec{E_1} = (E_x, E_y, E_z).$$

$$\vec{B_0} = B_0 \hat{z} \qquad \vec{k} = k\hat{x}$$

$$\vec{v_{e1}} = v_x \hat{x} + v_y \hat{y}.$$

$$i\vec{k}\,(i\vec{k}\cdot\vec{E_1}) - (ik\,ik\,\vec{E_1}) = -(-i\omega)\left[\frac{1}{c^2}(-i\omega\vec{E}) + \mu_0 \vec{j_1}\right].$$

$E_1$ 中有 x 分量. $i\vec{k}(ikE_x\hat{x})$

化为: $$-k^2 E_x \hat{x} + k^2 \vec{E_1} = +\frac{\omega^2}{c^2}\vec{E_1} + i\omega\mu_0 \vec{j_1}.$$

$$\vec{j_1} = -en_0\vec{v_{e1}}$$

动力学方程:

$$m\frac{\partial}{\partial t}\vec{v_{e1}} = -e\left[\vec{E_1} + \vec{v_{e1}} \times \vec{B_0}\right].$$

分量:

$$\begin{cases} m(-i\omega)v_x = -eE_x - ev_y B_0 \\ m(-i\omega)v_y = ev_x B_0 - eE_y \end{cases}$$

解得:

$$v_y = \frac{-ie}{\omega m}(E_y - v_x B_0).$$

$$v_x = \frac{-ie}{m\omega}\left[E_x + \frac{B_0(-ie)}{m\omega}(E_y - v_x B_0)\right]$$

$$= \frac{-ie}{m\omega}E_x - \frac{Be^2}{m^2\omega^2}E_y + \frac{Be^2}{m^2\omega^2}v_x$$

$$\Rightarrow \quad v_x = (1-\omega_c^2/\omega^2)^{-1}\frac{e}{m\omega}\left(-iE_x - \frac{\omega_c}{\omega}E_y\right).$$

$$v_y = (1-\omega_c^2/\omega^2)^{-1}\frac{e}{m\omega}\left(-iE_y + \frac{\omega_c}{\omega}E_x\right).$$

由 $-k^2 E_x \hat{x} + k^2 \vec{E}_1 = +\frac{\omega^2}{c^2}\vec{E}_1 + i\omega\mu_0 \vec{j}_1.$

$$\omega^2 \vec{E}_1 - \frac{i\omega e n_0}{\varepsilon_0}\frac{e}{m\omega}\left(-iE_x - \frac{\omega_c}{\omega}E_y\right)(1-\frac{\omega_c^2}{\omega^2})^{-1} = 0.$$

$$(1-\frac{\omega_c^2}{\omega^2})\omega^2 E_x - i\omega_p^2\left(-iE_x - \frac{\omega_c}{\omega}E_y\right) = 0$$

$$\left[(1-\omega_c^2/\omega^2)\omega^2 - \omega_p^2\right]E_x + i\frac{\omega_c\omega_p^2}{\omega}E_y = 0.$$

$$k^2 E_y = \frac{\omega^2}{c^2}E_y + i\omega\mu_0(-en_0)v_y.$$

$$\Rightarrow \quad c^2k^2 E_y = \omega^2 E_y - \frac{i\omega\, en_0}{\varepsilon_0}\frac{e}{m\omega}\left(-iE_y + \frac{\omega_c}{\omega}E_x\right)(1-\omega_c^2/\omega^2)^{-1}$$

$$(c^2k^2-\omega^2)E_y(1-\frac{\omega_c^2}{\omega^2}) = -\omega_p^2 E_y - \frac{i\omega_c}{\omega}\omega_p^2 E_x.$$

$$\left[(c^2k^2-\omega^2)(1-\frac{\omega_c^2}{\omega^2}) + \omega_p^2\right]E_y + \frac{\omega_c}{\omega}\omega_p^2 iE_x = 0.$$

$\det|A| = 0$

$$= -\left(\frac{\omega_c}{\omega}\omega_p^2\right)^2 - \left[(1-\omega_c^2/\omega^2)\omega^2 - \omega_p^2\right]\left[(c^2k^2-\omega^2)(1-\frac{\omega_c^2}{\omega^2}) + \omega_p^2\right]$$

$$\Leftrightarrow \quad c^2k^2\left(1-\frac{\omega_c^2}{\omega^2}\right)(\omega^2-\omega_h^2)=(\omega^2-\omega_h^2)^2-\frac{\omega_c^2}{\omega^2}\omega_p^4.$$

$$c^2k^2=\frac{(\omega^2-\omega_h^2)^2-(\omega_c/\omega)^2\omega_p^4}{(1-\omega_c^2/\omega^2)(\omega^2-\omega_h^2)}.$$

$$\frac{c^2k^2}{\omega^2}=\frac{(\omega^2-\omega_h^2)^2-(\omega_c/\omega)^2\omega_p^4}{(\omega^2-\omega_c^2)(\omega^2-\omega_h^2)}.$$

$$=\frac{(\omega^2-\omega_h^2)(\omega^2-\omega_c^2-\omega_p^2)-(\omega_c/\omega)^2\omega_p^4}{(\omega^2-\omega_c^2)(\omega^2-\omega_h^2)}.$$

$$=\frac{(\omega^2-\omega_h^2)(\omega^2-\omega_c^2)-\omega_p^2(\omega^2-\omega_h^2)-(\omega_c/\omega)^2\omega_p^4}{(\omega^2-\omega_c^2)(\omega^2-\omega_h^2)}.$$

$$=1+\frac{-\omega_p^2(\omega^2-\omega_h^2)-(\omega_c/\omega)^2\omega_p^4}{(\omega^2-\omega_c^2)(\omega^2-\omega_h^2)}.$$

$$=1-\frac{\omega_p^2}{\omega^2}\frac{\omega^2(\omega^2-\omega_h^2)+\omega_c^2\omega_p^2}{(\omega^2-\omega_c^2)(\omega^2-\omega_h^2)}.$$

$$=1-\frac{\omega_p^2}{\omega^2}\frac{\omega^2(\omega^2-\omega_c^2)-\omega^2\omega_p^2+\omega_c^2\omega_p^2}{(\omega^2-\omega_c^2)(\omega^2-\omega_h^2)}$$

$$=1-\frac{\omega_p^2}{\omega^2}\frac{\omega^2-\omega_p^2}{\omega^2-\omega_h^2}$$

$\Leftrightarrow$ 非寻常波色散关系：

$$c^2k^2(\omega^2-\omega_h^2)=\omega^2(\omega^2-\omega_h^2)-\omega_p^2(\omega^2-\omega_p^2)$$

① $k \to \infty$. $\omega = 0$ or $\omega = \omega_h$.

② $k = 0$